

7 (a) charge is quantised / discrete quantities B1 [1]

(b) (i) parallel so that the electric field is uniform / constant
horizontal so that *either* oil drop will not drift sideways

or field is vertical

or electric force is equal to weight B1 [2]

(ii) $qE = mg$

$$q \times 850 / (5.4 \times 10^{-3}) = 7.7 \times 10^{-15} \times 9.8$$

$$q = 4.8 \times 10^{-19} \text{ C } \underline{\text{and is negative}}$$

C1

C1

A1 [3]

(c) charge changes by $1.6 \times 10^{-19} \text{ C}$ between droplets / integral multiples
so charge on electron is $1.6 \times 10^{-19} \text{ C}$

M1

A0 [1]

8 (a) since momentum before combining is zero

B1